Abhinav Raj

Integrated M.Sc. Computer Science (AI/ML)

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PROFESSIONAL SUMMARY

Integrated M.Sc. Computer Science (AI/ML) graduate with a strong foundation in Machine Learning, Deep Learning, Computer Vision, and biosignal processing. UGC-NET qualified (95.31 percentile, Assistant Professor & Ph.D. eligible) with hands-on experience designing end-to-end AI systems using Python, TensorFlow, PyTorch, Django, and Streamlit. Experienced in building real-time machine learning pipelines, conducting AI research, and developing full-stack intelligent applications, with strong interest in Generative AI, intelligent automation, and production-ready AI systems.

EDUCATION

Integrated M.Sc. in Computer Science with Specialization in AI and ML

Nehru Arts and Science College, Kanhangad | Kannur University | 2021–2026

First Class with Distinction | OGPA: 8.803/10 (88.03%)

ACHIEVEMENTS

Qualified UGC-NET DECEMBER 2025 (Computer Science and Applications)

University Grants Commission (NTA) | E-Certificate Number: 25D/03/031644 | Percentile: 95.31

PROJECTS

FrugalMyo: Real-Time Myoelectric Gesture Recognition using Deep Learning

Academic Project, Tenth Semester – 2026

Developed a real-time sEMG gesture recognition system using a custom EMG acquisition setup and deep learning.

- Designed and trained a lightweight 1D CNN achieving ~87% classification accuracy for hand gesture classification.
- Designed and implemented an end-to-end machine learning pipeline covering signal acquisition, preprocessing, segmentation, model training, evaluation, and real-time inference.
- Developed a Streamlit interface for live EMG visualization, gesture prediction, confidence estimation, and real-time inference.
- Research work currently being extended for publication.

Tech Stack: Python, TensorFlow, Streamlit, NumPy, SciPy

Replication of NINAPRO DB1 Myoelectric Benchmark

Academic Project, Ninth Semester – 2025 | GitHub: github.com/ARJ010/ninapro-db1-benchmark-replication

Replicated the landmark 2015 Atzori et al. benchmark on the NINAPRO DB1 dataset.

- Designed a memory-efficient two-pass numpy.memmap pipeline capable of processing high-dimensional EMG features on an 8 GB RAM system.
- Reproduced benchmark findings identifying MAV + MLP as the optimal feature-classifier combination for real-time prosthetic control.
- Optimized classifier implementations, significantly reducing training time while maintaining benchmark performance.

Tech Stack: Python, NumPy, SciPy, scikit-learn

Attendance Tracking System (ATS) | Deployed & In Use at Nehru Arts and Science College

Independent Institutional Project – 2024 | *GitHub: github.com/ARJ010/Attendance_Tracking_System*

Designed and developed a Django-based attendance management system for Kerala's Four-Year Undergraduate Programme (FYUGP).

- Automated attendance logging and CSV report generation through an instructor-friendly workflow.
- Designed an optimized relational database schema for efficient student, course, and attendance management.
- Deployed and maintained in production, currently supporting 99 faculty members, 748+ students, 325 course papers, and 53,000+ attendance records.

Tech Stack: Django, MySQL, Bootstrap

Checklist Management System (CMS) | Deployed & In Use at TH&Co Chartered Accountants

Academic Project, Sixth Semester – 2023 | *GitHub: github.com/ARJ010/checklist_app*

Developed a role-based workflow management system to streamline audit and procedural operations for a Chartered Accountant firm.

- *Implemented secure multi-role access (Admin, User, Checker) with checklist automation, revision tracking, and workflow management.*
- *Enabled task management, form tracking, and audit workflow coordination through a centralized web application.*
- *Deployed and continuously used in production for over 3 years, supporting day-to-day audit and compliance workflows.*

Tech Stack: Django, SQLite, HTML, CSS, JavaScript

SKILLS

- **Programming Languages:** Python, SQL, Java, JavaScript, C, HTML, CSS, LaTeX
- **AI & Machine Learning:** TensorFlow, PyTorch, scikit-learn, NumPy, SciPy, Pandas, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), Biosignal Processing (sEMG)
- **Web & Full-Stack Development:** Django, React, Streamlit
- **Databases:** MySQL, SQLite
- **Tools & Platforms:** Git, GitHub, Jupyter Notebook, VS Code
- **Core Competencies:** Data Engineering, Software Engineering, Database Design, Research Methodology, Technical Writing, Problem Solving, Critical Thinking

CERTIFICATIONS

TensorFlow Developer Professional Certificate

DeepLearning.AI – May 2025

Credential ID: UABTJPA585N8

- Completed full specialization covering CNNs, NLP, and Time Series Forecasting using TensorFlow 2.x
- Included hands-on projects and model implementation in real-world AI tasks

Mathematical Foundations for Machine Learning (Elite Certification)

NPTEL & IISc Bangalore – Oct 2025

Credential ID: NPTEL25CS136S959800072

- Awarded "Elite" status for performance in the 12-week advanced curriculum covering Linear Algebra, Probability, and Optimization

Project Completion Certificate – Checklist Management System

TH&Co Chartered Accountant Firm – Jul 2024

- Recognized for successful deployment and implementation of CMS

Web Development Basics (HTML, CSS, JavaScript)

University of Michigan – Nov 2023

- Credential IDs: F26NYDE6TBYC, 9NB9NJ6JV9L7, GGDU78ZXCLE7

Programming for Everybody (Getting Started with Python)

University of Michigan – Nov 2021

Credential ID: P2XYRZ3W3F96

- Covered foundational programming concepts in Python